thumbnail gallery of 15 word clouds derived from text in a species biodiversity database. Oftentimes, word clouds rotate the words to set them at right angles or jumble them at random angles to improve the appearance of the graphics. Word clouds, like streamgraphs, receive criticism for being hard to read or presenting too little information. But both are surprisingly popular with audiences.

I created these word clouds using my data with the popular Java applet Wordle, which provides an easy UI and a few aesthetic customization choices. Wordle has flooded the internet with word clouds because it lets anyone create visually arresting but problematic graphics by dropping text onto a page. This caused much consternation among data visualization experts, who think word clouds are evil because they embed no analysis in the visualization and only highlight superficial data such as the quantity of words in a blog post.

But word clouds aren't evil. First of all, they're popular with audiences. But more than that, words are remarkably effective graphical objects. If you can identify a numerical attribute that indicates the significance of a word, then scaling the size of a word in a word cloud relays that significance to your reader.

So let's start by assuming we have the right kind of data for a word cloud. Fortunately, we do: the top twenty words used in this chapter, with the number of each word.

**Listing 5.9 worddata.csv**

```
text,frequency
layout,63
function,61
data,47
return,36
attr,29
chart,28
array,24
style,24
layouts,22
values,22
need,21
nodes,21
pie,21
use,21
figure,20
circle,19
we'll,19
zoom,19
append,17
elements,17
```

To create a word cloud with D3, you have to use another layout that isn't in the core library, created by Jason Davies (who created the sentence trees using the tree layout shown in figure 5.17). You'll also need to implement an algorithm written by Jonathan Feinberg (http://static.mrfeinberg.com/bv_ch03.pdf). The layout, `d3.layout.cloud()`, is available on GitHub at https://github.com/jasondavies/d3-cloud. It requires that